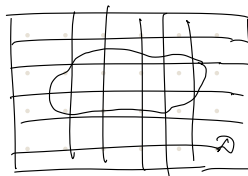
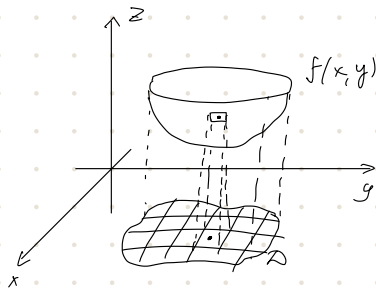
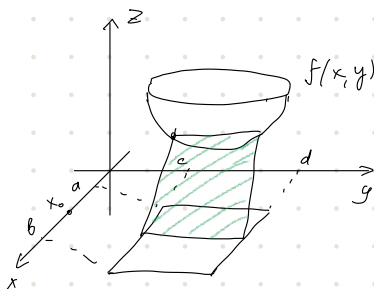


16.10.25

6 пр. Интегралы



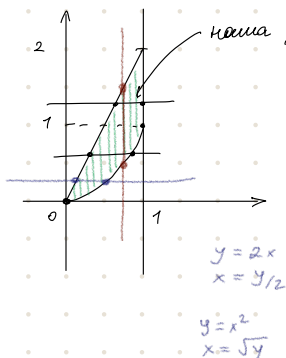
$\iint_D f(x, y) dx dy$ — проитер. по этой области



① Свести двойной интеграл

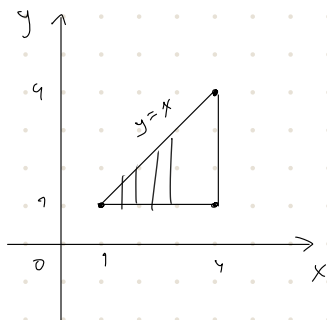
$\iint_D f(x, y) dx dy$ к повторному двойня способами

а) $D: x=1, y=x^2, y=2x (x \leq 1)$



$$\begin{aligned} \iint_D f(x, y) dx dy &= \int_0^1 dx \int_{x^2}^{2x} f(x, y) dy = \\ &= \int_0^1 dy \int_{y/2}^{\sqrt{y}} f(x, y) dx + \int_1^2 dy \int_{\frac{y}{2}}^1 f(x, y) dx \end{aligned}$$

д) D-треугольн. с вершин. $(1, 1), (4, 1), (4, 4)$



$$\iint_D f(x, y) dx dy = \int_1^4 dx \int_1^x f(x, y) dy =$$

$$= \int_1^4 dy \int_y^4 f(x, y) dx$$

б) D-Δ: $(2, 1), (5, 2), (3, 7)$

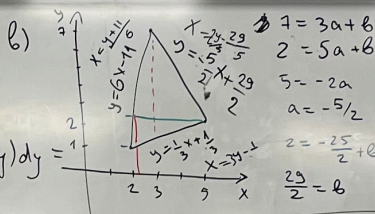
$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1}$$

$$y = ax + b$$

$$\iint_D f(x, y) dx dy =$$

$$= \int_2^3 dx \int_{\frac{1}{3}x + \frac{1}{3}}^{6x-11} f(x, y) dy + \int_3^5 dx \int_{\frac{1}{3}x + \frac{1}{3}}^{\frac{5}{2}x + \frac{29}{2}} f(x, y) dy$$

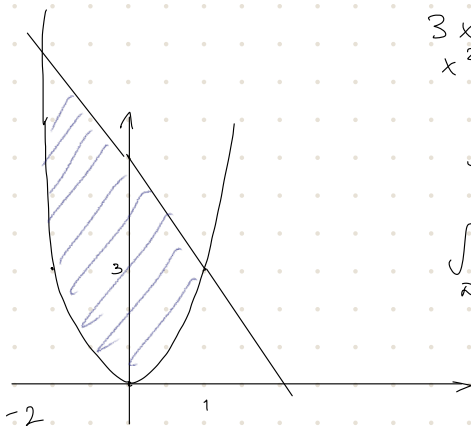
$$= \int_1^2 dy \int_{\frac{y+11}{6}}^{3y-1} f(x, y) dx + \int_2^7 dy \int_{\frac{y+11}{6}}^{\frac{2y-29}{5}} f(x, y) dx$$



$$\begin{cases} y = ax + b \\ 7 = 3a + b \\ 1 = a + 2b \\ 6 = a \\ 1 = 6 + b \\ -11 = -b \end{cases}$$

$$\begin{cases} y = 6x - 11 \\ 7 = 3a + b \\ 2 = 5a + b \\ 5 = -2a \\ a = -5/2 \\ 2 = -25/2 + b \\ 25/2 = b \end{cases}$$

2) D: $y = 3x^2, y = 6 - 3x$



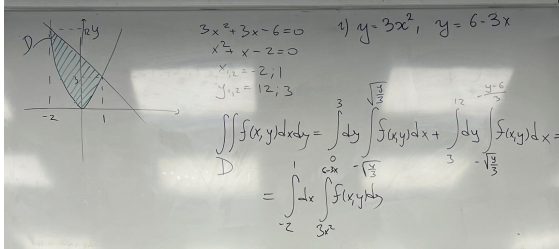
$$3x^2 = 6 - 3x$$

$$x^2 + x - 2 = 0$$

$$x = 1 \quad x = -2$$

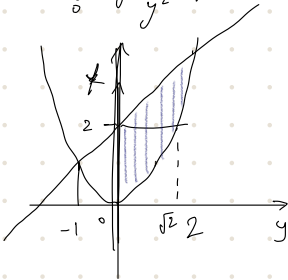
$$y = 3 \quad y = 12$$

$$\iint_D f(x, y) dx dy = \int dy \int f(x, y) dx$$

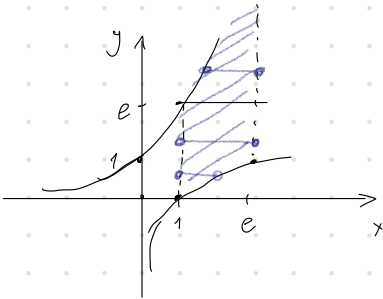


② Записать в виде повторного, изменяя порядок интер

a) $\int_0^2 dy \int_{y^2}^{y+2} f dx = \int_0^2 dx \int_0^{\sqrt{x}} f dy + \int_2^4 dx \int_{x-y}^{\sqrt{x}} f dy$

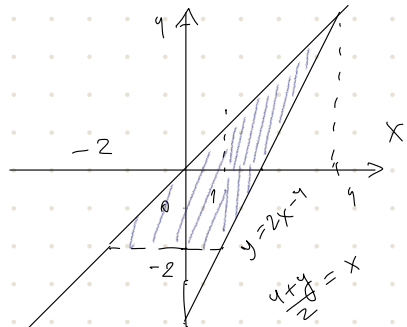


d) $\int_1^e dx \int_{\ln x}^{e^x} f dy = \int_0^1 dy \int_1^{e^y} f dx + \int_1^e dy \int_1^{e^y} f dx + \int_e^{e^e} dy \int_{\ln y}^{e^y} f dx$



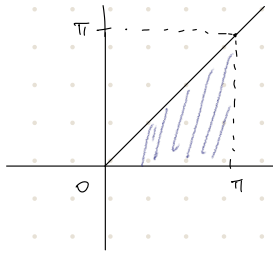
$y = \ln x$
 $e^y = e^{\ln x} = x$

b) $\int_{-2}^1 dx \int_{-2}^x f dy + \int_1^4 dx \int_{2x-4}^x f dy =$
 $= \int_{-2}^4 dy \int_{\frac{y+2}{2}}^{\frac{y+4}{2}} f dx$



③ Выведем:

$$a) \int_0^{\pi} dy \int_y^{\pi} \frac{\sin x}{x} dx = \int_0^{\pi} dx \int_0^x \frac{\sin x}{x} dy = \int_0^{\pi} \frac{\sin x}{x} \cdot y \Big|_0^x dx =$$



$$= \int_0^{\pi} \frac{\sin x}{x} x dx = \int_0^{\pi} \sin x dx = -\cos x \Big|_0^{\pi} =$$

$$= -(-1 - 1) = \boxed{2}$$

2/3: 1) вывести $\iint_D f dx dy$ к норм. 2-м-м

a) D: $1 \leq (x-2)^2 + y^2 \leq 4$

b) D: $\triangle : (0, 0), (0, 3), (-2, 0), (-1, 0)$

2) норм. м.

$$\int_0^1 dx \int_0^{\sqrt{1-x^2}} (1-y^2)^{3/2} dy$$

$$b) \int_0^1 dx \int_x^1 \sqrt{1-y^2} dy$$